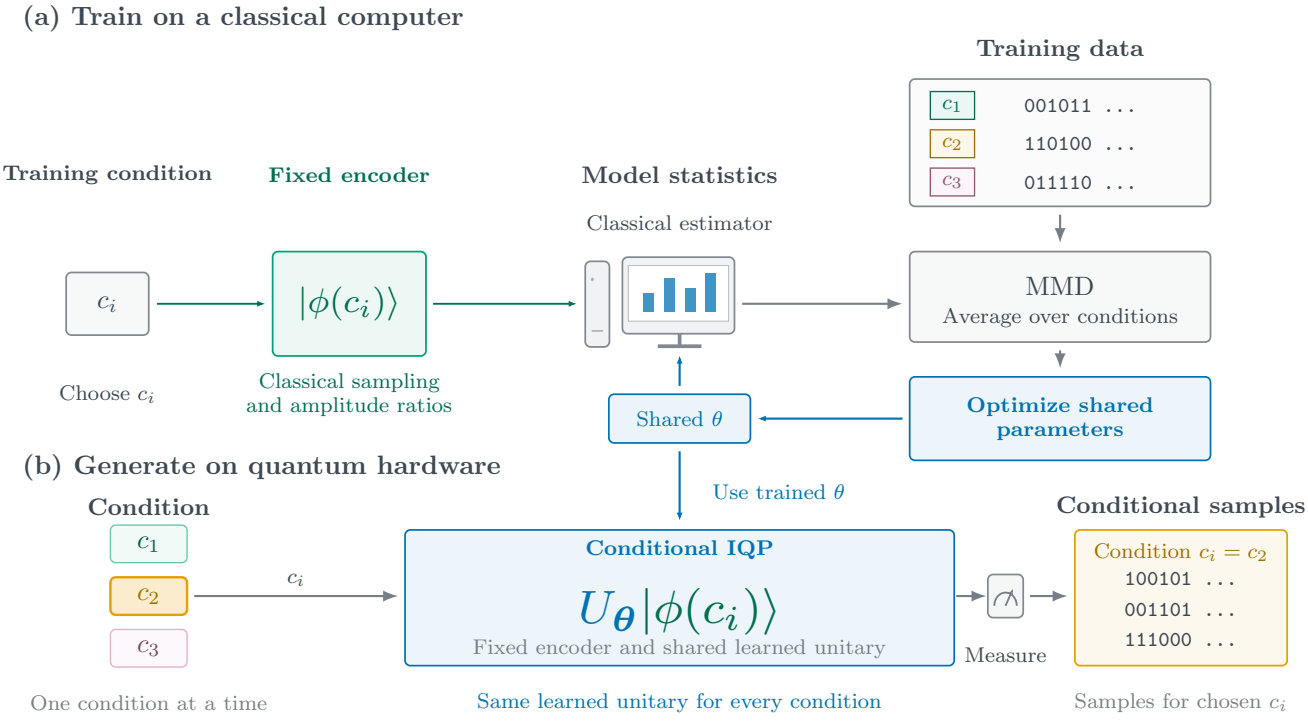

(a) Train on a classical computer
Training data
c_1 001011 ...
c_2 110100 ...
c_3 011110 ...
Training condition
Fixed encoder
Model statistics
Classical estimator
c_i
$|\phi(c_i)\rangle$
MMD
Average over conditions
Choose c_i
Classical sampling and amplitude ratios
Shared θ
Optimize shared parameters
(b) Generate on quantum hardware
Use trained θ
Condition
c_1
c_2
c_3
c_i
Conditional IQP
$U_{\boldsymbol{\theta}}|\phi(c_i)\rangle$
Fixed encoder and shared learned unitary
Measure
Conditional samples
Condition $c_i = c_2$
100101 ...
001101 ...
111000 ...
One condition at a time
Same learned unitary for every condition
Samples for chosen c_i